
EML Trees Are Universal Approximators

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Abstract

1 The recently introduced EML (Exp-Minus-Log) function acts as continuous ana-
2 logue of NAND gates, providing a compositional building block capable of rep-
3 resenting elementary functions. In this work, we study the expressive power of
4 tree-structured compositions of EML functions. We show that such trees enjoy
5 a universal approximation property for functions in $W^{k,\infty}$ for $k \in \mathbb{N}$, drawing
6 on classical neural network approximation arguments while exploiting the abil-
7 ity to explicitly construct EML trees that mimic polynomial representations. We
8 further propose a learning algorithm for EML-type trees equipped with fitting
9 parameters, and demonstrate its feasibility in practical optimization problems. Our
10 results establish EML trees as a theoretically grounded framework for function
11 approximation.

12 1 Introduction

$$\text{EML}(x, y) = e^x - \text{Log } y \quad (1)$$

13 Analogy to NAND gates

14 Mention that EML is a mathematical curiosity, more than a new paradigm for function approximation

15 State that since we have all elementary functions, that we can recognize that we should expect an
16 approximation theorem, since we can build polynomial

17 2 Background

18 EQL, KAN, AI-Feynman, SINDy (/ PDEFind), Symbolic regression, SymTorch, Bacon (Langley)

19 Symbolic metamodel:

- 20 • Demystifying Black-box Models with Symbolic Metamodels (NeurIPS)

21 3 Universal Approximation Theorem

22 For our work, we consider a generalization of the EML function (1), denoted EML_{θ_i} , to include 6
23 learnable parameters each, and represented as

$$\text{EML}_{\theta_i}(x, y) = a_i e^{b_i x + c_i} + d_i \ln(e_i y + f_i), \quad (2)$$

24 where i enumerates the instance of the EML function in the tree and

$$\theta_i = \{a_i, b_i, c_i, d_i, e_i, f_i\}. \quad (3)$$

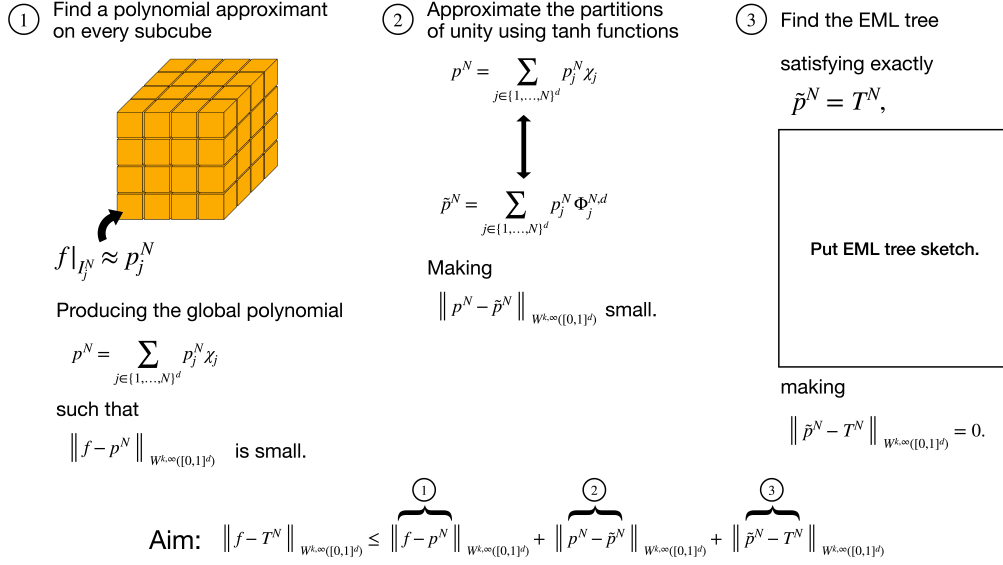


Figure 1: Sketch of the proof of the universal approximation theorem. **THIS IS AN OLDER SCHEMATIC; NEEDS UPDATE.**

25 For ..., we will often refer to individual EML_{θ_i} as “atoms” and the reusable functions or operations
 26 built using a collection of EML’s with fixed parameters as “blocks”.

27 Of course, one observes that setting $a_i = b_i = e_i = 1$, $d_i = -1$, and $c_i = f_i = 0$ recovers the original
 28 EML function (1).

29 The purposes behind this generalization are both technical and practical.

30 As will be detailed in the upcoming subsections, proving the approximation theorem relies on
 31 exactly representing a polynomial as an EML tree, but the technical lemma 2, which allows us to
 32 approximate the original function by a polynomial does not give us an estimate or an upper bound on
 33 the coefficients of these polynomials; thus, with the current framework, it is impossible to know the
 34 size of the EML tree that represents such polynomials, because in the original function (1), constants
 35 need to build recursively from smaller ones (starting from 1), so reaching coefficients badly depends
 36 on their magnitude.

37 Additionally, practically, as we will witness when we construct operations using these generalized
 38 EML functions, they are able to represent operations and functions more compactly, as will be
 39 demonstrated for the basic binary operations $(+, -, \times, \div)$ and others used in building polynomials.

40 It also performs better.

41 Another advantage is that we now no longer need to consider the number 1 as a possible input to the
 42 EML atoms, thus reducing the inputs to the EML atoms as either the variables or other EML atoms.

43 3.1 Setting

44 3.2 Statement of the Theorem

45 **Theorem 1** (Universal Approximation Theorem in $W^{k,\infty}$).

46 3.3 Outline of the Proof

47 3.4 Remarks

48 **Remark 1.** For non positive functions.

49 **Remark 2.** For recovering trigonometric functions.

50 **Remark 3.** *There is an inherent nonuniqueness associated with these generalized EMLs. (comment*
 51 *on addition block for example).*

52 **Remark 4.** *Corollary to any domain via affine transformation.*

53 **Remark 5.** *Removing the need for approximating indicator functions by adding ReLU function to*
 54 *definition of EML (?).*

55 4 Implementation

56 The constructive proof of Theorem 1 establishes that the parametric EML grammar can express,
 57 exactly, any element of an arbitrarily fine polynomial approximant to a $W^{k,\infty}$ function. It does not
 58 by itself address whether such a tree can be *recovered* from data by gradient-based optimisation.
 59 In contrast to Odrzywolek [2026], which targets *symbolic* recovery of vanilla-EML trees over the
 60 discrete grammar $\{1, x, \text{EML}(\cdot, \cdot)\}$ and reports rapid collapse of recovery rate at depth ≥ 5 , we
 61 exploit the continuous parameters of the generalised atom $\text{EML}_{\theta_i}(x, y)$ to make the optimisation
 62 landscape amenable to first-order methods. This section reports a small, focused empirical study in
 63 support of the theorem’s practical content.

64 4.1 Setup

65 **Architecture.** We use the PEM tree of Section 1 directly: a complete binary tree T_θ of $2^d - 1$
 66 atoms, each carrying six free parameters $\theta_i = \{a_i, b_i, c_i, d_i, e_i, f_i\}$ and computing $\text{EML}_{\theta_i}(x, y)$.
 67 Children are fed *raw* between atoms, with the input variable x entering at the leaf-parents directly.
 68 For univariate x the parameter count is $\#\theta(d) = 6(2^d - 1) \in \{42, 90, 186\}$ at depths $d \in \{3, 4, 5\}$.

69 **Two parameterisations.** We consider two regimes:

70 **Real (R).** All $\theta_i \in \mathbb{R}^6$. To keep the operator real on $\mathbb{R}$, the logarithm’s argument is routed through
 71 $\ln(\text{softplus}(\cdot) + \epsilon)$ with $\epsilon = 10^{-6}$, a smoothed-positive surrogate for the principal-branch
 72 logarithm. This blocks the trigonometric path of the corresponding remark in Section 1 but
 73 eliminates branch-cut pathologies.

74 **Complex (C).** All $\theta_i \in \mathbb{C}^6$, stored as twin real Parameters (θ_i^R, θ_i^I) . Forward in $\mathbb{C}^{128}$ with the
 75 principal branch of $\ln$; the loss is computed on $\text{Re}(T_\theta(x))$. We add a small auxiliary penalty
 76 $\lambda \cdot |\text{Im } T_\theta|^2$ with $\lambda = 10^{-3}$ so that the trained predictor is honestly real-valued. Doubling the
 77 parameter count is the cost; gaining the trigonometric path is the intended benefit.

78 **Optimisation procedure.** We minimise mean-squared error on a uniform grid of $N = 200$ samples
 79 per target with the following two-stage strategy, applied per (target, depth, parameterisation) cell at a
 80 fixed wall-clock budget of 30 seconds:

- 81 1. **Adam multi-restart.** From independent Gaussian-initialised seeds, run 1500 steps of Adam with
 82 learning rate 5×10^{-3} , cosine-annealed to zero, gradient ℓ_2 -clip at 5.0. Track the best mean-squared
 83 error across restarts.
- 84 2. **L-BFGS refinement.** Apply L-BFGS with strong-Wolfe line search (≤ 200 iterations, history 20)
 85 to the best-of-restarts model and take the smaller of the pre- and post-refinement losses.

86 The number of Adam restarts per cell is determined by the budget rather than fixed and is reported in
 87 Table 1.

88 **Assumptions.** The reported numbers depend on three explicit choices. (i) Under R the operator is
 89 the softplus-stabilised log surrogate, not vanilla EML; the two coincide on positive arguments and
 90 diverge smoothly near zero. (ii) All targets are noise-free, dense samples on the stated domain; no
 91 claim is made about generalisation to sparse or noisy data. (iii) The wall-clock budget is small (30 s on
 92 a single CPU) and held fixed across cells; reported errors at the larger depths are optimisation-limited,
 93 not capacity-limited.

94 4.2 Targets and metric

95 We report results on five 1D targets, each chosen to align with a specific construction in the proof:

Table 1: Best relative RMSE per cell, PEM trees with real (R) and complex (C) parameters at depths 3, 4, 5. Strategy: Adam multi-restart + L-BFGS, 30 s wall-clock per cell. Parenthesised integer is the number of completed Adam restarts within the budget; parameter count is $\#\theta = 6(2^d - 1)$ for R and twice that for C. Bolded entry per row is the best of the six.

Target	Real (R), $\#\theta = 42, 90, 186$			Complex (C), $\#\theta = 84, 180, 372$		
	$d=3$	$d=4$	$d=5$	$d=3$	$d=4$	$d=5$
$x^3 - x$	$7.58 \cdot 10^{-3}$ (15)	$1.47 \cdot 10^{-3}$ (8)	$4.32 \cdot 10^{-3}$ (4)	$2.17 \cdot 10^{-2}$ (7)	$1.37 \cdot 10^{-2}$ (4)	$5.20 \cdot 10^{-2}$ (2)
$\tanh(2x)$	$4.20 \cdot 10^{-3}$ (15)	$1.22 \cdot 10^{-3}$ (8)	$2.10 \cdot 10^{-3}$ (4)	$1.77 \cdot 10^{-3}$ (7)	$5.16 \cdot 10^{-3}$ (4)	$1.80 \cdot 10^{-2}$ (2)
$\sin(x)$	$3.02 \cdot 10^{-3}$ (15)	$7.67 \cdot 10^{-3}$ (8)	$1.01 \cdot 10^{-2}$ (4)	$1.74 \cdot 10^{-2}$ (7)	$3.49 \cdot 10^{-3}$ (4)	$2.07 \cdot 10^{-2}$ (2)
e^{-x^2}	$4.13 \cdot 10^{-3}$ (15)	$1.12 \cdot 10^{-3}$ (7)	$4.66 \cdot 10^{-3}$ (4)	$2.20 \cdot 10^{-2}$ (7)	$5.27 \cdot 10^{-3}$ (3)	$1.27 \cdot 10^{-2}$ (2)
$\sin(3x) e^{-x^2/2}$	$3.23 \cdot 10^{-2}$ (15)	$9.80 \cdot 10^{-2}$ (7)	$9.09 \cdot 10^{-2}$ (4)	$6.45 \cdot 10^{-2}$ (6)	$3.91 \cdot 10^{-2}$ (3)	$3.57 \cdot 10^{-2}$ (2)

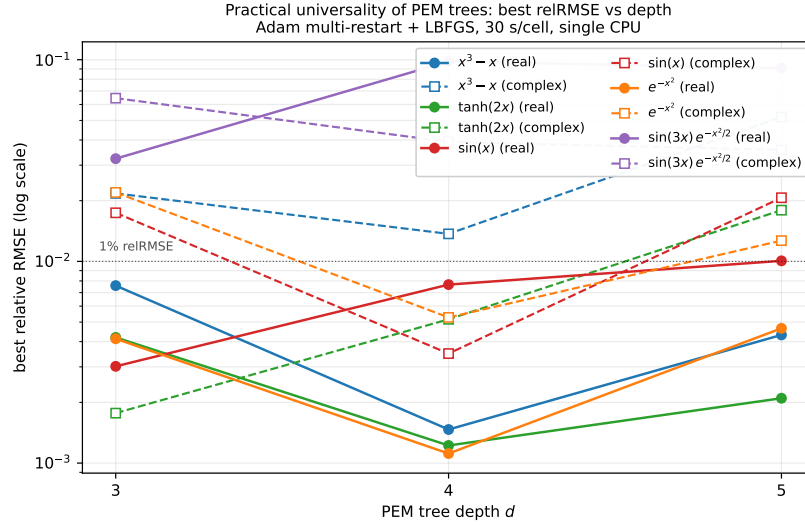


Figure 2: Best relative RMSE vs PEM tree depth, five targets, real (solid) and complex (dashed) parameterisations, 30 s wall-clock per cell. The dotted line marks the 1% relRMSE threshold. Real PEM reaches sub-percent error on 4/5 targets by depth 4; complex PEM is competitive only on $\sin(3x) e^{-x^2/2}$, where its larger capacity helps despite the halved restart count.

- $x^3 - x$ on $[-1.5, 1.5]$: a degree-3 polynomial; the local approximant of Lemma 2 in its simplest non-trivial form. The constructive bound is $5k - 3 = 12$ atoms for degree $k = 3$.
- $\tanh(2x)$ on $[-2, 2]$: hyperbolic tangent, used in the partition-of-unity construction (Section 6); the explicit EML realisation is in Figure 5.
- $\sin(x)$ on $[-\pi, \pi]$: the trigonometric case highlighted in the corresponding remark; the vanilla-EML construction routes through $\ln(-1) = i\pi$, unreachable from R.
- e^{-x^2} on $[-2.5, 2.5]$: smooth analytic, no oscillation — baseline for “easy” targets.
- $\sin(3x) e^{-x^2/2}$ on $[-3, 3]$: oscillation $\times$ envelope; the hardest target in this set.

We report *relative RMSE*, $\text{relRMSE} = \|T_\theta(x) - f(x)\|_2 / \|f(x)\|_2$, on the training grid, best-of-restarts.

4.3 Results

Table 1 reports the best relRMSE per cell. Figure 2 plots relRMSE against depth on a log scale.

What the numbers say. Three observations support the theorem’s practical content. (i) Real-parameterised PEM trees achieve sub-percent relRMSE on 4 of 5 targets at depth 4 within 30 s of

single-CPU compute, including the polynomial target $x^3 - x$ which falls to 1.47×10^{-3} . This is a direct empirical reflection of the constructive route of the theorem (polynomial approximation, then exact EML representation): gradient methods reach close to the constructed optimum, even though the construction itself is not used as an initialisation. (ii) The trigonometric target $\sin(x)$ does *not* require the complex parameterisation in our setting: real PEM at depth 3 already reaches 3.0×10^{-3} . The softplus-stabilised log is expressive enough on this domain that the complex path is not the binding constraint at the budget shown. (iii) The hardest target, $\sin(3x) e^{-x^2/2}$, is the one clear case where the complex parameterisation pays off: 3.6×10^{-2} at depth 5 (C) versus 9.1×10^{-2} at the same depth (R). The composite oscillation-envelope target rewards the extra capacity even though C completes only half as many restarts in the same wall-clock budget.

The general pattern visible in Figure 2 — error falling sharply from depth 3 to depth 4, then rising at depth 5 for several real targets — is a wall-clock-budget effect: deeper trees have more parameters but, at fixed budget, fewer restarts to find good basins. With unlimited compute this artefact would disappear; under fixed compute it suggests an “elbow” depth around $d = 4$ for these 1D targets.

Computational cost. The full sweep (5 targets $\times$ 3 depths $\times$ 2 parameterisations = 30 cells) completes in approximately 12 minutes on a single laptop CPU (Apple M-series, no GPU). Per-cell wall-clock is 30 s by construction; the number of completed Adam restarts ranges from ~ 15 at ($d=3$, R) down to 2 at ($d=5$, C). No GPU acceleration or parallelism was used.

4.4 Limitations

We list the limitations of this empirical study honestly, in priority order.

- **Single-run-of-experiment.** Tabled values are the best loss from one run of the multi-restart procedure. Variance across full reruns is not reported. Order-of-magnitude conclusions are robust; comparisons within a factor of ~ 2 should not be over-interpreted.
- **Real-softplus surrogate.** The R parameterisation is not vanilla EML. Trees fit under R coincide with EML on the positive-argument slice but smoothly diverge near zero. The complex parameterisation uses the operator literally.
- **Univariate, smooth, dense, noise-free.** Multivariate fitting, sparse sampling, and noise robustness are all open and likely harder; the theorem is dimension-agnostic but our experiments are not.
- **Fixed-budget effects on depth comparison.** Deeper trees receive fewer restarts at fixed wall-clock, so the depth-vs-error curve confounds capacity gain with restart-count loss. The dip at depth 4 for several targets is the visible signature.
- **Optimisation, not approximation, theory.** The reported numbers are achieved errors, not bounds. They are upper-bounded by the theorem; whether they meet the rate the theorem predicts (and at what budget the gap closes) is a question we do not address here.
- **Function-level fits, not symbolic recoveries.** Decoding the fitted depth-3 trees and greedy-snapping each parameter to the nearest of $\{-2, -1, 0, 1, 2\}$ at 30% relative-loss tolerance, we recover at most 7/42 (17%) parameters as discrete on any target, and zero atoms become fully discrete. On $\tanh(2x)$ and e^{-x^2} exactly 0 parameters are snappable. The optimiser exploits the $\mathbb{R}^6$ freedom of each atom to land in a continuous-coefficient basin whose function value matches the target but whose parameters bear no resemblance to the constructive symbolic blocks of Sections 5–6. Recovering interpretable trees would require a structural regulariser (sparsity-toward- $\{0, \pm 1\}$, or warm-start from the construction); this is left for future work.

5 Discussion

- Generalization to other spaces (Besov)
- Remark on analytic functions
- Finding a way to use the vanilla EMLs by following another proof technique that involves finding the best polynomial approximant with bounded coefficients and finding upper bounds for constructing these constants (which is a challenging task to get the most optimal upper bounds for getting high constants).

- 159 • We could maybe make due with less than 6 parameters per EML atom.
- 160 • Trying to find better techniques to optimize these trees.
- 161 • Probably shorter trees exist that do the same thing; but, in this case, they were constructed
- 162 like this to be human readable and easy to verify and interpretable.
- 163 • Can we find another function that trades depth for “width“, by resorting to making the atoms
- 164 take in more inputs (similar to network constructions)
- 165 • Practically, the logarithm takes in only positive values, so we could put an absolute value
- 166 inside the logarithm to circumvent any potential issue with domain
- 167 • Discussion when talking about possible use for interpretable scientific discovery (Wyder
- 168 et al. [2025], Yermakov et al. [2025])
- 169 **Open Question 1.** *Can we*

170 A Notation

171 $\mathbb{N}^d$

172 $\times$ denotes cartesian product

173 B Sobolev Spaces

174 We recall the definitions of some basic functional spaces.

175 The Lebesgue spaces, denoted $L^p(\Omega)$ are defined as

$$L^p(\Omega) = \left\{ f : \Omega \rightarrow \mathbb{R} \mid \int_{\Omega} |f(x)|^p dx < \infty \right\}, \quad (4)$$

176 for $1 \leq p < \infty$, equipped with the norm

$$\|f\|_{L^p(\Omega)} = \left(\int_{\Omega} |f(x)|^p dx \right)^{\frac{1}{p}}. \quad (5)$$

177 When $p = \infty$, where $L^\infty(\Omega)$ consists of functions that are essentially bounded and whose norm is

$$\|f\|_{L^\infty(\Omega)} = \operatorname{ess\,sup}_{x \in \Omega} |f(x)| = \inf \{ C \geq 0 \mid |f(x)| \leq C \text{ for almost every } x \in \Omega \}. \quad (6)$$

178 Moreover, the Sobolev space $W^{k,p}$ is defined as

$$W^{k,p}(\Omega) = \left\{ f \in L^p(\Omega) \mid D^\alpha f \in L^p(\Omega) \text{ for all } \alpha \in \mathbb{N}_0^d \text{ with } |\alpha| \leq k \right\}. \quad (7)$$

179 We also define associated seminorms. For $p < \infty$, we have

$$|f|_{W^{m,p}(\Omega)} = \left(\sum_{|\alpha|=m} \|D^\alpha f\|_{L^p(\Omega)}^p \right)^{\frac{1}{p}}, \quad \text{for } m = 0, \dots, k, \quad (8)$$

180 and for $p = \infty$, we have

$$|f|_{W^{m,\infty}(\Omega)} = \max_{|\alpha|=m} \|D^\alpha f\|_{L^\infty(\Omega)}, \quad \text{for } m = 0, \dots, k. \quad (9)$$

181 Based on these seminorms, we can define the following norms for $p < \infty$,

$$\|f\|_{W^{k,p}(\Omega)} = \left(\sum_{m=0}^k |f|_{W^{m,p}(\Omega)}^p \right)^{\frac{1}{p}}, \quad (10)$$

182 and for $p = \infty$,

$$\|f\|_{W^{k,\infty}(\Omega)} = \max_{0 \leq m \leq k} |f|_{W^{m,\infty}(\Omega)}. \quad (11)$$

183 In particular, when $p = 2$, the Sobolev spaces $H^k(\Omega) = W^{k,2}(\Omega)$ (with $k \in \mathbb{N}_0$) are Hilbert spaces
 184 with the corresponding norms $\|\cdot\|_{H^k(\Omega)} = \|\cdot\|_{W^{k,2}(\Omega)}$ and seminorms $|\cdot|_{H^m} = |\cdot|_{W^{m,2}}$ for any integer
 185 m such that $0 \leq m \leq k$.

186 Additionally, the space of k -times continuously differentiable functions $C^k(\Omega)$, with the associated
 187 norm $\|f\|_{C^k(\Omega)} = \|f\|_{W^{k,\infty}(\Omega)}$.

188 We will make frequent use of the following Banach algebra property.

189 **Lemma 1.** *Let $d \in \mathbb{N}$, $k \in \mathbb{N}_0$, and $\Omega \subset \mathbb{R}^d$. If $f, g \in W^{k,\infty}(\Omega)$, then*

$$\|fg\|_{W^{k,\infty}(\Omega)} \leq 2^k \|f\|_{W^{k,\infty}(\Omega)} \|g\|_{W^{k,\infty}(\Omega)}. \quad (12)$$

190 C Useful Definitions and Lemmas

191 We recall the definition of a star-shaped set, which is a necessary geometric requirement for the
 192 approximation of functions by polynomials in Sobolev spaces (in Lemma 2).

193 **Definition 1.** A set $A \subset \mathbb{R}^d$ is star-shaped with respect to every point in a subset $B \subset A$ if

$$\forall y \in B, \forall x \in A, [y, x] \subset A, \quad (13)$$

194 where $[y, x] = \{(1-t)y + tx \mid t \in [0, 1]\}$ is the straight line in $\mathbb{R}^d$ connecting the point y to the point
195 x .

196 We recall the Bramble-Hilbert lemma for Sobolev spaces, as proved in De Ryck et al. [2021], with
197 similar statements appearing in Durán [1983] and Verfürth [1999].

198 **Lemma 2.** Let $\Omega \subset \mathbb{R}^d$ be an open and bounded set of diameter $0 < h < e^{-1/2}d^{-3/2}$ which is
199 star-shaped with respect to every point in an open ball $B \subset \Omega$ with diameter ph . Then, for every
200 $f \in W^{s,\infty}(\Omega)$, there exists a polynomial $\hat{f}$ of degree at most $s-1$ such that for any $k \in \mathbb{N}_0$ with $k < s$,
201 it holds that

$$\|f - \hat{f}\|_{W^{k,\infty}(\Omega)} \leq \frac{\pi^{1/4} \sqrt{s} (d\sqrt{de}h)^{s-k}}{(s-k-1)!} |f|_{W^{s,\infty}(\Omega)}. \quad (14)$$

202 D Required Building Blocks

203 In this section, we present the explicit constructions of the basic binary operations, exponentiation,
204 and the hyperbolic tangent function, all of which are used in the approximation proof (Theorem 1).

205 D.1 Basic Binary Operations

206 In Figure 3, we provide the EML trees (with their associated parameters) to produce the basic binary
207 operations of addition, subtraction, multiplication, and division. The addition and subtraction blocks
208 are used in the constructions of the multiplication and division blocks respectively.

209 As observed in Figure 3, we note that the size and depth requirements of these binary operations are
summarized in Table 2.

Operator	$N(\cdot)$	Depth($\cdot$)
$x +_{\theta} y$	3	2
$x -_{\theta} y$	3	2
$x \times_{\theta} y$	6	4
$x \div_{\theta} y$	6	4
$x_{\theta}^n, mx_{\theta}^n$	2	2
$\tanh_{\theta}(x), m \tanh_{\theta}(x)$	8	5

Table 2: Size and depth of the operations in Appendix D.

210

211 D.2 Exponentiation

212 In our proof, we will also make frequent use of exponentiation to a constant power. We provide
213 explicit constructions for EML trees that perform x^n and mx^n for constants $m, n \in \mathbb{R}$ in Figure 4.

214 Note that in Figure 4b, we construct mx^n directly, which we use to represent the monomial terms of
215 polynomials (in Appendix E). This is because it is more efficient to incorporate the coefficient within
216 the exponentiation block itself, rather than introducing a separate multiplication block to combine a
217 constant with the x^n block.

218 For both exponentiation blocks (4a and 4b), the size and depth of the associated EML trees are 2 (and
219 appended to Table 2).

220 D.3 Hyperbolic Tangent

221 The hyperbolic tangent is constructed, using the exponential definition, by first getting $e^{2x} - 1$ and
222 $e^{2x} + 1$ (via two separate EML blocks) and then dividing them, as in Figure 5. The size of the tanh
223 block is 8 and its depth is 5, which are appended to Table 2.

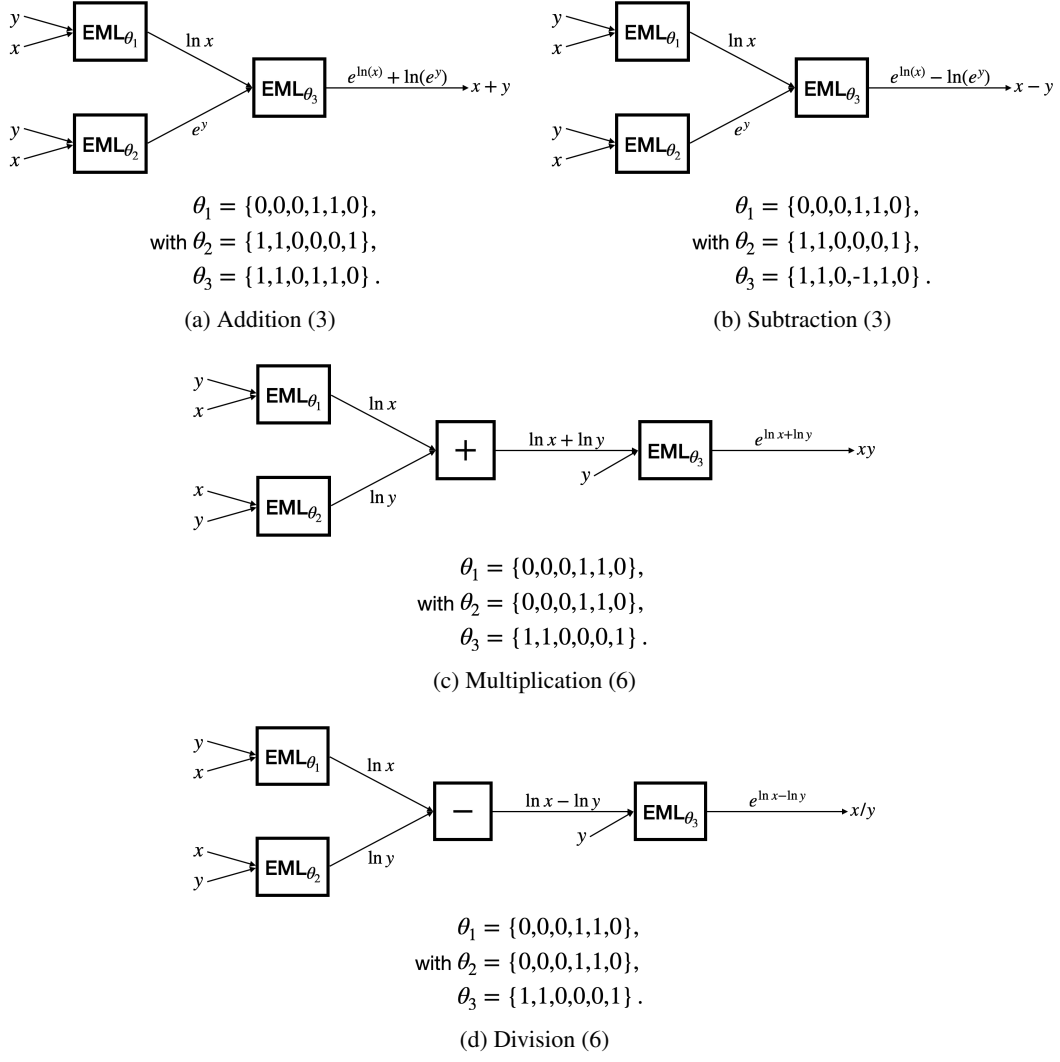


Figure 3: The Basic Binary Operations

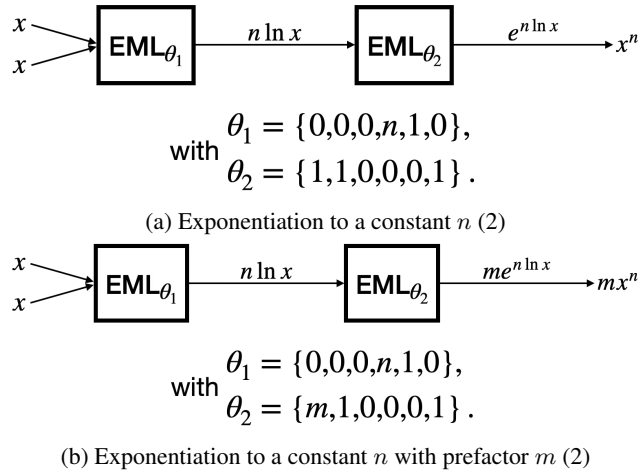


Figure 4: Exponentiating

224 We can similarly obtain $m \tanh x$, which is useful for the construction of the approximate partitions
 225 of unity, by amending the division block to include the prefactor m . This is done by changing θ_3 in
 the division block (Figure 3d) to be $\theta_3 = \{m, 1, 0, 0, 1\}$.

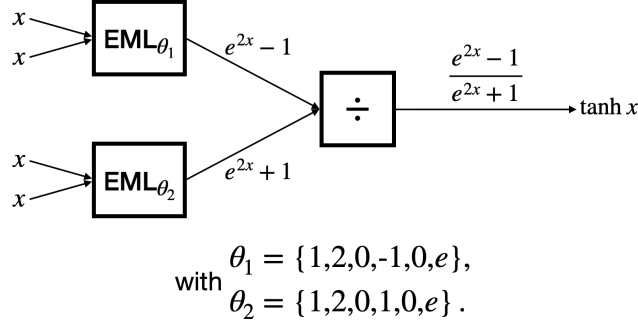


Figure 5: Hyperbolic Tangent (8)

226

227 E Constructing Polynomials

228 The following two subsections detail how to construct univariate and multivariate polynomials using
 229 EML trees and provide upper bounds for the number of required EML blocks in each case.

230 E.1 Univariate Polynomials

231 A univariate polynomial of degree k , denoted by

$$p(x) = \sum_{i=0}^k a_i x^i = a_0 + a_1 x + a_2 x^2 + \cdots + a_k x^k, \quad (15)$$

232 can be (naively) constructed using EML blocks as in Figure 6, wherein we used two simple constituent
 233 blocks: addition (Figure 3a) and exponentiation to a constant with a prefactor (Figure 4b).

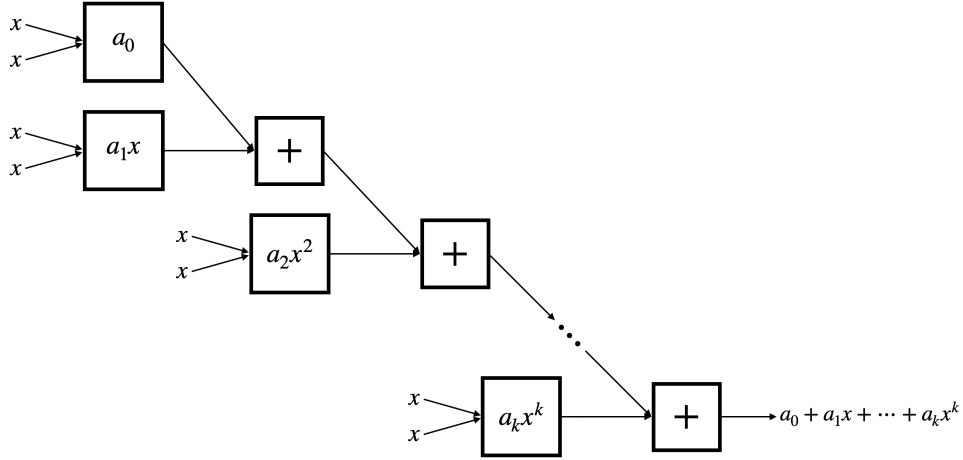


Figure 6: Univariate Polynomial (5s - 3)

234 The upper bound on the number of required EML blocks is obtained by noting that we used

- 235 • $k - 1$ addition blocks (each requiring 3 EML atoms), and
- 236 • k exponentiation with a prefactor blocks (each requiring 2 EML atoms).

237 So, constructing an EML tree p_θ such that $p_\theta = p$ (as per the above procedure) requires $N(T_\theta) =$
 238 $3(k - 1) + 2k = 5k - 3$ and $\text{Depth}(T_\theta) = 2 + 2(k - 1) = 2k$, since the longest path starts with the x
 239 feeding into the a_0 block and passes through all $k - 1$ addition blocks.

Hence, we phrase the above conclusion as the lemma:

Lemma 3 (Construction of Univariate Polynomials). *Given a univariate polynomial p of degree k , there exists an EML tree p_θ (following the explicit construction in Figure 6), such that $p_\theta = p$ with $N(p_\theta) \leq 5k - 3$ and $\text{Depth}(p_\theta) \leq 2k$.*

E.2 Multivariate Polynomials

We recall that, for $\mathbf{x} = (x_1, \dots, x_d) \in \mathbb{R}^d$, a multivariate polynomial p of degree k is of the form

$$p(\mathbf{x}) = \sum_{|\alpha| \leq k} c_\alpha x_1^{\alpha_1} x_2^{\alpha_2} \dots x_d^{\alpha_d}, \quad (16)$$

where $\alpha = (\alpha_1, \alpha_2, \dots, \alpha_d)$ is a multi-index with $|\alpha| = \alpha_1 + \alpha_2 + \dots + \alpha_d$, and c_α are the associated coefficients. Such a polynomial (of degree $\leq k$) has $\binom{d+k}{k}$ monomial terms.

To construct an EML tree for the polynomial (16), (1) we construct the monomials $c_\alpha x_1^{\alpha_1}, x_2^{\alpha_2}, \dots, x_d^{\alpha_d}$ for each α with $|\alpha| \leq k$, (2) multiply these monomials together (separately for each α), (3) do steps (1) and (2) for every monomial, and (4) add them together. A breakdown of the required EML tree sizes and depths for each step is as follows:

- (1) For a certain α , each monomial, of which there are d , requires an exponentiation block (and the first has a prefactor), totaling $2d$ EML atoms;
- (2) Again for each α , we need to multiply the d monomials, requiring $d - 1$ multiplication blocks and, thus, $6(d - 1)$ EML atoms; thus, each term, considering both getting monomials and multiplying them requires $2d + 6(d - 1) = 8d - 6$ EML atoms;
- (3) Since we have $\binom{d+k}{d}$ terms in the polynomial, we do steps (1) and (2) $\binom{d+k}{d}$ times, thus requiring $(8d - 6)\binom{d+k}{d}$ EML atoms;
- (4) We finally sum over all terms, requiring $\binom{d+k}{d} - 1$ addition blocks, each consisting of 3 EML atoms.

Thus, in total, constructing the polynomial p in (16) takes

$$(8d - 6)\binom{d+k}{d} + 3\left(\binom{d+k}{d} - 1\right) = (8d - 3)\binom{d+k}{d} - 3 \quad (17)$$

EML atoms, and whose depth can be calculated (by drawing a schematic like Figure 6) to be

$$2 + 4(d - 1) + 2\left(\binom{d+k}{d} - 1\right) = 4d + 2\binom{d+k}{d} - 4, \quad (18)$$

since the x 's feed into the first $c_\alpha x^{\alpha_1}$ block (of depth 2), pass through $d - 1$ multiplication (each of depth 4), and then through $\binom{d+k}{d} - 1$ addition blocks (each of depth 2).

The above justified the following lemma:

Lemma 4 (Construction of Multivariate Polynomials). *Given $p : \mathbb{R}^d \rightarrow \mathbb{R}$, a multivariate polynomial of degree k , there exists an EML tree p_θ , such that $p_\theta = p$ with*

$$N(p_\theta) \leq (8d - 3)\binom{d+k}{d} - 3, \quad \text{and} \quad \text{Depth}(p_\theta) \leq 4d + 2\binom{d+k}{d} - 4. \quad (19)$$

F Approximating the partition of unity

In this section, we aim to construct an approximate partition of unity, similar to Section 4 of De Ryck et al. [2021]. We do this because indicator functions are unsuitable to be represented by EML trees due to the fact that they contain a discontinuity and cannot be suitably represented by chaining elementary functions constructed from compositions of EML. For this reason, we construct smooth approximations of indicator functions on subsets of $[0, 1]^d$ via tanh functions. When suitably shifted and parametrized, these functions output values close to 1 within the subset and close to 0 outside of it.

276 F.1 Properties of the approximate partition of unity

277 We first recall some of the properties of the partition of unity of De Ryck et al. [2021].

278 For a point $y \in \mathbb{R}$ and a parameter $\alpha \in \mathbb{R}$ (to be chosen later), we define the one-dimensional
279 approximate indicator functions as

$$\begin{aligned}\rho_1^N(y) &= \frac{1}{2} - \frac{1}{2}\sigma\left(\alpha\left(y - \frac{1}{N}\right)\right), \\ \rho_m^N(y) &= \frac{1}{2}\sigma\left(\alpha\left(y - \frac{m-1}{N}\right)\right) - \frac{1}{2}\sigma\left(\alpha\left(y - \frac{m}{N}\right)\right) \quad \text{for } 2 \leq m \leq N-1, \\ \rho_N^N(y) &= \frac{1}{2}\sigma\left(\alpha\left(y - \frac{N-1}{N}\right)\right) + \frac{1}{2}.\end{aligned}\tag{20}$$

280 We can clearly see that if α is large enough, these act as soft indicator functions over the intervals
281 $\left(\frac{m-1}{N}, \frac{m}{N}\right)$ (depending on the subscript of ρ).

282 To choose α , we note that since $\sigma(x) \rightarrow 1$ as $x \rightarrow \infty$ and its derivatives also saturate ($\sigma^{(m)}(x) \rightarrow 0$
283 with $x \rightarrow \infty$), we can choose $R > 0$ large enough such that $|\sigma^{(m)}|$ is decreasing over $[R, \infty)$ for all
284 $1 \leq m \leq k$.

285 So, given $\varepsilon > 0$, we choose α , depending on N and ε , large enough such that

$$\frac{\alpha}{N} \geq R, \quad 1 - \sigma\left(\frac{\alpha}{N}\right) \leq \varepsilon, \quad \alpha^m \left| \sigma^{(m)}\left(\frac{\alpha}{N}\right) \right| \leq \varepsilon \text{ for all } 1 \leq m \leq k,\tag{21}$$

286 justified also by Lemma A.4 in De Ryck et al. [2021]. Consequently, α can be chosen to satisfy the
287 above conditions and has the explicit value

$$\alpha = N \max\left\{R, \ln\left(\frac{(2k)^{k+1}(Nk)^k}{e^k \varepsilon}\right)\right\},\tag{22}$$

288 (see Lemma A.5 in De Ryck et al. [2021] for justification).

289 Importantly, we define

$$\Phi_j^N(x) = \prod_{i=1}^d \rho_{j_i}^N(x_i),\tag{23}$$

290 which acts as our approximate partition of unity. We also define the set $\mathcal{V}_d$ as

$$\mathcal{V}_d = \left\{v \in \mathbb{Z}^d \mid \max_{1 \leq i \leq d} |v_i| \leq 1\right\},\tag{24}$$

291 which allow us choose only the cubes directly surrounding a cube of interest.

292 Lemmas 5 and 6 below prove that the set of functions $\{\Phi_j^N\}_{j \in \mathcal{A}^N}$ acts as an approximate partition of
293 unity in the sense that they satisfy the following two properties: for every $j \in \mathcal{A}^N$,

$$\sum_{v \in \mathcal{V}_d} \Phi_{j+v}^{N,d} \approx 1 \quad \text{and} \quad \sum_{\substack{v \notin \mathcal{V}_d, \\ j+v \in \mathcal{A}^N}} \Phi_{j+v}^{N,d} \approx 0.\tag{25}$$

294 We recall the below two lemmas from De Ryck et al. [2021]. The first provides a quantitative bound
295 for how close to 1 is the contribution of Φ_j^N and its direct neighbors on I_j^N .

296 **Lemma 5** (Lemma 4.1 of De Ryck et al. [2021]). *If $0 < \varepsilon < 1/4$, then*

$$\left\| \sum_{v \in \mathcal{V}_d} \Phi_{j+v}^N - 1 \right\|_{W^{k,\infty}(I_j^N)} \leq 2^{dk} d\varepsilon.\tag{26}$$

297 The second lemma guarantees that the contribution of the functions far away from the cube I_j^N is
298 small.

299 **Lemma 6** (Lemma 4.2 of De Ryck et al. [2021]). *Let $k \in \mathbb{N}_0$ and $v \in \mathbb{Z}^d$ with $v_i \geq 2$ for all $1 \leq i \leq N$.
300 Then, it holds that*

$$\|\Phi_{j+v}^N\|_{W^{k,\infty}(I_j^N)} \leq \max\{1, (2k)^{2k} \alpha^k\} \varepsilon.\tag{27}$$

301 F.2 Constructing the approximate partition of unity using EML trees

302 In Figure 7, we construct the ρ_m^N functions for $2 \leq m \leq N-1$, according to the definition in (20), and provide the associated parameters.

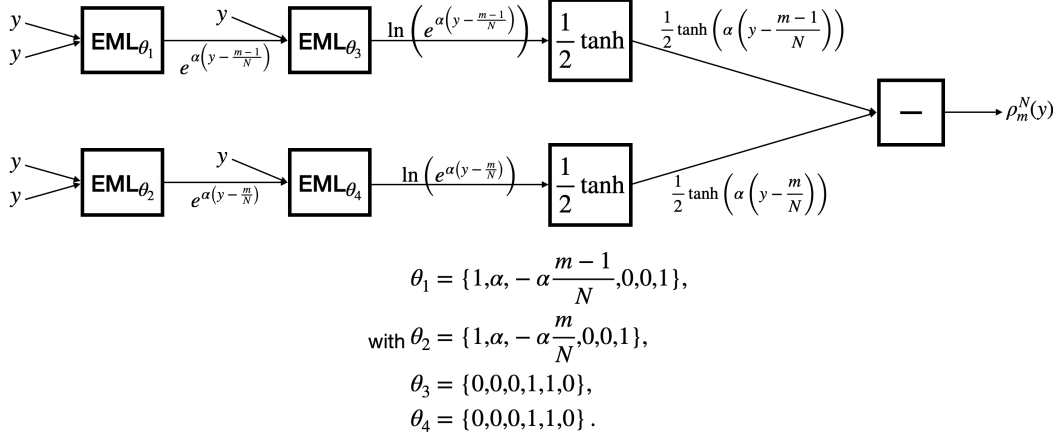


Figure 7: Construction of ρ_m^N for $2 \leq m \leq N-1$

303

304 This construction requires 4 EML blocks, 2 tanh blocks with $1/2$ constants, and a subtraction block,
305 which boils down to a size of $4 + 2 \times 8 + 3 = 23$ and a depth of $1 + 1 + 5 + 2 = 9$.

306 ρ_1^N and ρ_N^N (from (20)) can be constructed more simply, requiring 3 EML blocks (one of which to
307 get the constant $1/2$), one tanh block, and one subtraction block, thus corresponding to a size of
308 $3 + 8 + 3 = 14$ and depth of $1 + 1 + 5 + 2 = 9$. But, for the sake of simplicity, we will consider all
309 instances of ρ_m^N used to be those corresponding to Figure 7.

310 Then, we construct $\Phi_j^{N,d}$, using (23) by multiplying d instances of the one-dimensional partitions,
311 $\rho_{j_i}^N$. This requires $d-1$ multiplication blocks, and thus the EML tree corresponding to $\Phi_j^{N,d}$ has size
312 $23d + 6(d-1) = 29d - 6$ and depth $9 + 4(d-1) = 4d + 5$, yielding the following lemma:

313 **Lemma 7** (Construction of Approximate Partitions of Unity). *For $N, d \in \mathbb{N}$ and $j \in \mathcal{A}^N$, there exists*
314 *an EML tree T_θ such that $T_\theta = \Phi_j^N$ with*

$$N(T_\theta) \leq 29d - 6, \quad \text{and} \quad \text{Depth}(T_\theta) \leq 4d + 5. \quad (28)$$

315 G Proof of Theorem 1

316 *Proof.* We set out to perform the steps for the proof as described in the main text above and in Figure
317 1.

318 **Step 1. Approximation by Local Polynomials.** We divide the domain $[0, 1]^d$ into cubes of side
319 length $1/N$. To denote these cubes, we use the corresponding index set

$$\mathcal{A}^N = \{1, \dots, N\}^d \quad (29)$$

320 of cardinality $|\mathcal{A}^N| = N^d$. For every $j \in \mathcal{A}^N$, we associate the cube defined as

$$I_j^N = \bigotimes_{i=1}^d \left(\frac{j_i - 1}{N}, \frac{j_i}{N} \right). \quad (30)$$

321 We also define overlapping cubes, which will be used to satisfy the condition for the Bramble-Hilbert
322 lemma, for the given N and every $j \in \mathcal{A}^N$ as

$$J_j^N = \bigotimes_{i=1}^d \left(\frac{j_i - 2}{N}, \frac{j_i + 1}{N} \right), \quad (31)$$

which is of diameter $\frac{3\sqrt{d}}{N}$. We notice that there exists a ball of diameter $3/N$ such that J_j^N is star-shaped with respect to every point in this ball (per Definition 1). Thus, for every $j \in \mathcal{A}^N$, by the Bramble-Hilbert lemma (Lemma 2), there exists a polynomial p_j^N of degree at most $s-1$ such that, for all $0 \leq k \leq s-1$

$$\begin{aligned} \|f - p_j^N\|_{W^{k,\infty}(J_j^N)} &\leq \frac{\pi^{1/4}\sqrt{s}}{(s-k-1)!} \left(\frac{5d^2}{N}\right)^{s-k} |f|_{W^{s,\infty}([0,1]^d)} \\ &\leq \max_{0 \leq m \leq k} \frac{\pi^{1/4}\sqrt{s}(5d^2)^{s-m}}{(s-m-1)!} \frac{|f|_{W^{s,\infty}([0,1]^d)}}{N^{s-k}} := \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}}, \end{aligned} \quad (32)$$

choosing $N > 5d^2$ (to obtain decay) and using $3\sqrt{e} \leq 5$.

328

Remark 6. *The polynomials are chosen to be good approximators on the overlapping cubes (not on the disjoint ones) because*

Step 2. Definition of the Global Polynomial. We then define the global piecewise polynomial over the entire domain $[0, 1]^d$ by

$$p^N = \sum_{j \in \mathcal{A}^N} p_j^N \chi_j, \quad (33)$$

where χ_j denotes the indicator function on I_j^N . Since p^N is composed using indicator functions (which are discontinuous), it cannot be adequately controlled in $W^{k,\infty}$ spaces where $k \geq 1$. This is why we introduce

$$\tilde{p}^N = \sum_{j \in \mathcal{A}^N} p_j^N \Phi_j^N, \quad (34)$$

as the global polynomial using the approximate partitions of unity and which we will ultimately represent using an EML tree. Similarly, we define $\tilde{f}^N$ as

$$\tilde{f}^N = \sum_{j \in \mathcal{A}^N} f \Phi_j^N. \quad (35)$$

Step 3. Approximation of f by $\tilde{f}^N$. We first prove that $\tilde{f}^N$ is close to f . Considering an arbitrary $i \in \mathcal{A}^N$ and any $k \geq 1$, we employ the property 1, Lemmas 5 and 6, and the explicit value of α (from (22)) to get

$$\begin{aligned} \|f - \tilde{f}^N\|_{W^{k,\infty}(I_i^N)} &= \left\| f - \sum_{j \in \mathcal{A}^N} f \Phi_j^N \right\|_{W^{k,\infty}(I_i^N)} \\ &\leq 2^k \|f\|_{W^{k,\infty}(I_i^N)} \left\| 1 - \sum_{v \in \mathcal{V}_d} \Phi_{i+v}^N \right\|_{W^{k,\infty}(I_i^N)} + 2^k \|f\|_{W^{k,\infty}(I_i^N)} \left\| \sum_{\substack{j \in \mathcal{A}^N \\ j-i \notin \mathcal{V}_d}} \Phi_j^N \right\|_{W^{k,\infty}(I_i^N)} \\ &\leq 2^k \|f\|_{W^{k,\infty}(I_i^N)} (2^{kd} d\varepsilon + N^d (2k)^{2k} \alpha^k \varepsilon) \\ &\leq 2^k \|f\|_{W^{k,\infty}(I_i^N)} \left(2^{kd} d + N^d (2k)^{2k} N^k (k+1)^k \max \left\{ R^k, \ln^k \left(\frac{2Nk^2}{\varepsilon^{\frac{1}{k+1}} e} \right) \right\} \right) \varepsilon \\ &\leq \delta (2(k+1))^{3k} \max \left\{ R^k, \ln^k \left(\frac{2Nk^2}{\varepsilon^{\frac{1}{k+1}} e} \right) \right\} \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}}, \end{aligned} \quad (36)$$

with ε chosen to satisfy

$$\varepsilon \leq \frac{\delta \mathcal{C}(d, k, s, f)}{2^{(k+1)d} d N^{s+d} \|f\|_{W^{k,\infty}([0,1]^d)}}. \quad (37)$$

Similarly, for $k = 0$, we have

$$\|f - \tilde{f}^N\|_{L^\infty(I_i^N)} \leq \|f\|_{L^\infty(I_i^N)} (d\varepsilon + N^d \varepsilon) \leq \frac{\delta \mathcal{C}(d, k, s, f)}{3 N^{s-k}}. \quad (38)$$

343 **Step 4. Approximation of $\tilde{f}$ by the Global Polynomial $\tilde{p}^N$.** For $k = 0$,

$$\begin{aligned}
\|\tilde{f}^N - \tilde{p}^N\|_{L^\infty(I_i^N)} &= \left\| \sum_{j \in \mathcal{A}^N} (f - p_j^N) \Phi_j^N \right\|_{L^\infty(I_i^N)} \\
&= \left\| \sum_{v \in \mathcal{V}_d} (f - p_{i+v}^N) \Phi_{i+v}^N \right\|_{L^\infty(I_i^N)} + \sum_{\substack{j \in \mathcal{A}^N \\ j-i \notin \mathcal{V}_d}} \|f - p_j^N\|_{L^\infty(I_i^N)} \|\Phi_j^N\|_{L^\infty(I_i^N)} \\
&\leq \left(\max_{v \in \mathcal{V}_d} \|f - p_{i+v}^N\|_{L^\infty(I_i^N)} \right) \left\| \sum_{v \in \mathcal{V}_d} \Phi_{i+v}^N \right\|_{L^\infty(I_i^N)} + \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}} N^d \varepsilon \\
&\leq \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}} (1 + d\varepsilon) + \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}} N^d \varepsilon \\
&\leq \left(1 + \frac{\delta}{3}\right) \frac{\mathcal{C}(d, k, s, f)}{N^{s-k}},
\end{aligned} \tag{39}$$

344 with suitably chosen ε .

345 Whereas for $k \geq 1$,

$$\begin{aligned}
\|\tilde{f}^N - \tilde{p}^N\|_{W^{k, \infty}(I_i^N)} &= \left\| \sum_{j \in \mathcal{A}^N} (f - p_j^N) \Phi_j^N \right\|_{W^{k, \infty}(I_i^N)} \\
&= \left\| \sum_{v \in \mathcal{V}_d} (f - p_{i+v}^N) \Phi_{i+v}^N \right\|_{W^{k, \infty}(I_i^N)} + \sum_{\substack{j \in \mathcal{A}^N \\ j-i \notin \mathcal{V}_d}} 2^k \|f - p_j^N\|_{L^\infty(I_i^N)} \|\Phi_j^N\|_{W^{k, \infty}(I_i^N)} \\
&\leq \text{something} + N^d 2^k
\end{aligned} \tag{40}$$

346 **Step 5. Finding the EML Tree.** In this step, our objective is to construct an EML tree T^N such that
347 it is exactly equal to the approximate global polynomial, that is $T^N = \tilde{p}^N$.

348 **Step 6. Conclusion.** Finally,

$$\begin{aligned}
\|f - T^N\|_{W^{k, \infty}([0,1]^d)} &\leq \|f - \tilde{f}^N\|_{W^{k, \infty}([0,1]^d)} + \|\tilde{f}^N - \tilde{p}^N\|_{W^{k, \infty}([0,1]^d)} + \|\tilde{p}^N - T^N\|_{W^{k, \infty}([0,1]^d)} \\
&\leq \|f - \tilde{f}^N\|_{W^{k, \infty}([0,1]^d)} + \|\tilde{f}^N - \tilde{p}^N\|_{W^{k, \infty}([0,1]^d)} \\
&\leq
\end{aligned} \tag{41}$$

349 **Step 7. Upper Bound on the Size of the EML Tree.**

350

□

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